

A Social Compass: Guiding Self-Regulation in High-Arousal Moments with Ubiquitous Sensing

Moshi Qin*

moshi.qin@tuni.fi

TAUCHI Research Center, Faculty of Information
Technology and Communication Sciences
Tampere University
Tampere, Pirkanmaa, Finland

Devbrat Anuragi*

devbrat.anuragi@tuni.fi

TAUCHI Research Center, Faculty of Information
Technology and Communication Sciences
Tampere University
Tampere, Pirkanmaa, Finland

Abstract

Social interactions often trigger physiological stress responses that can impair communication effectiveness and well-being. This paper presents SocialCompass, a proof-of-concept framework that leverages the large multimodal foundation model, ImageBind, to detect high-stress social moments and provide timely self-regulation support. Unlike existing emotion recognition systems that focus on classification accuracy, SocialCompass prioritizes real-world deployment and user autonomy. Our core innovation lies in a novel zero-shot inference strategy that bridges multiple public datasets without requiring direct cross-dataset alignment. We train a social context classifier on high-frequency Inertial Measurement Unit (IMU) and audio features from the ExtraSensory dataset, while developing an arousal inference model that uses ImageBind's cross-modal semantic alignment to compare real-time sensor embeddings with conceptual text anchors derived from psychological concepts. This dual-model approach allows us to identify High-Arousal Social Events (HASE) in real-world settings. We propose a three-phase implementation strategy culminating in a demonstration on smart glasses. Initial feasibility analysis shows promising potential for real-world deployment while highlighting key technical and ethical considerations for socially-aware computing systems.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools; Human computer interaction (HCI);
• **Computing methodologies** → Machine learning.

Keywords

Multimodal sensing, stress recognition, zero-shot learning, social interaction, wearable computing, affective computing, emotion regulation

ACM Reference Format:

Moshi Qin and Devbrat Anuragi. 2025. A Social Compass: Guiding Self-Regulation in High-Arousal Moments with Ubiquitous Sensing. In *Companion of the the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo,

*Both authors contributed equally to this research.



This work is licensed under a Creative Commons Attribution 4.0 International License. *UbiComp Companion '25*, Espoo, Finland
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3750591>

Finland. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3714394.3750591>

1 Introduction and Motivation

Social interactions are fundamental to human experience, yet they often involve cognitive demands that can exceed our processing capabilities. Neuroscientific research has established that elevated physiological arousal during conversations can trigger limbic system dominance, leading to impulsive responses that individuals later regret. The phenomenon of emotional hijacking [4], where the amygdala triggers fight-or-flight responses during social stress, has been extensively documented in cognitive neuroscience literature. However, practical solutions for real-time detection and intervention using everyday technology remain largely unexplored.

Current emotion recognition systems in social contexts have three fundamental limitations restricting their practical utility. First, most existing approaches rely on single modalities—primarily facial expressions or speech analysis—which provide incomplete emotional assessment. While recent advances in wearable sensor technology have demonstrated automated stress detection during conversations using physiological and inertial sensors [1], and emerging research explores cross-modal approaches using multi-sensory wearables [3] and real-world smartwatch data [2], these multi-modal solutions often rely on training with small, lab-collected datasets and do not leverage the power of modern foundation models to bridge diverse, real-world data sources. Second, existing systems often require intrusive sensors or controlled laboratory environments, constraining real-world applicability. Third, current approaches prioritize emotion classification over actionable intervention, failing to translate recognition into meaningful behavioral support for users in social situations.

This paper presents SocialCompass, a novel framework that addresses these limitations through a practical, user-centered approach. Rather than pursuing perfect emotion classification, we focus on detecting moments when users might benefit from self-regulation support. Our primary contribution lies in our novel methodology for leveraging ImageBind's emergent zero-shot capabilities to bridge disparate public datasets without requiring direct cross-dataset alignment; we demonstrate how this approach enables the use of consumer-grade sensors for a system that delivers timely, discreet cognitive support through smart glasses.

Our study explores three research questions: (1) Can high-arousal social moments be detected reliably using only smartphone-based sensing and foundation model capabilities? (2) How can interventions be designed to support autonomy without being intrusive or

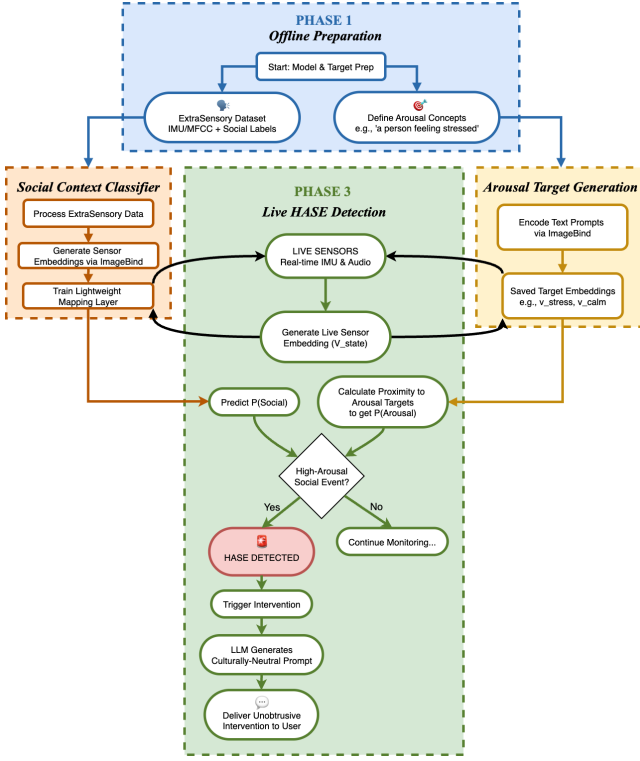


Figure 1: Architectural overview of the SocialCompass framework.

directive? (3) What trade-offs are required to ensure the feasibility of on-device deployment through knowledge distillation and model optimization?

2 Technical Approach

The SocialCompass framework employs a novel dual-model architecture to detect High-Arousal Social Events (HASE) through ubiquitous sensing. Our approach addresses the fundamental challenge of bridging disparate datasets while maintaining computational efficiency for real-world deployment on smart glasses. The implementation follows a three-phase strategy: offline preparation and model training, target embedding generation, and live demonstration pipeline.

2.1 Phase 1: Offline Preparation and Dataset Utilization

Our framework leverages two complementary public datasets for offline model training. The ExtraSensory dataset [6] serves as our primary source for real-world sensor data, providing tri-axial Inertial Measurement Unit (IMU) data sampled at 40 Hz, 13-dimensional Mel-Frequency Cepstral Coefficient (MFCC) audio features, and rich user-reported context labels such as ‘talking’ and ‘conversation’. However, it lacks direct arousal annotations. For validation of our zero-shot arousal inference approach, we utilize the GLOBEM [7] and WESAD datasets [5], which provide ground truth for arousal

states through PANAS scores (GLOBEM) and controlled stress induction protocols (WESAD). All sensor streams are segmented into 20-second windows with z-score normalization applied to IMU data to account for device variations. We exclude windows with more than 20% missing data to ensure quality.

2.2 Phase 2: Dual-Model Training and Target Generation

We model HASE detection as the probabilistic intersection of two independently developed models: a social context classifier (C_{social}) and an arousal inference model ($S_{arousal}$), where C denotes ‘classifier’ and S denotes ‘similarity-based inference’. This dual-model approach allows us to leverage the unique strengths of different datasets while avoiding complex cross-dataset alignment issues.

2.2.1 Social Context Classifier (C_{social}). The C_{social} model determines real-time social interaction engagement using ExtraSensory data. We generate binary ground-truth labels by mapping context annotations to social states, with ‘talking’, ‘conversation’, and ‘meeting’ as positive samples.

For each 20-second window, tri-axial IMU data is normalized and transformed into a time-series tensor, while we utilize the provided 13-dimensional MFCC audio features. Both modalities are passed through ImageBind’s respective sensor encoders to produce a unified latent representation V_{state} , which captures the user’s momentary physical state and vocal activity. A lightweight classifier is then trained to map these ImageBind sensor embeddings to the binary *is_social* ground-truth labels using cross-validation.

2.2.2 Arousal Target Generation ($S_{arousal}$). Our $S_{arousal}$ model addresses the critical limitation of lacking large-scale datasets with synchronized sensor data and arousal labels. We propose a novel zero-shot inference approach that leverages ImageBind’s cross-modal semantic alignment capabilities.

During the offline preparation phase, we define conceptual text prompts describing different arousal states (“a person feeling stressed and agitated” for high arousal, “a person feeling calm and relaxed” for low arousal). These prompts are processed through ImageBind’s text encoder to create static target embeddings (e.g., V_{stress} , V_{calm}) in the shared latent space, which are saved for real-time inference.

2.3 Phase 3: Live HASE Detection and Intervention

During real-time operation, live sensor data from smartphones or smart glasses is continuously processed through our detection pipeline. Real-time IMU and audio data is processed through ImageBind’s sensor encoders to generate the live sensor embedding V_{state} . This embedding is simultaneously fed to both the trained C_{social} classifier (producing probability $P(\text{Social})$) and compared with the pre-computed arousal target embeddings through cosine similarity to derive probability $P(\text{Arousal})$. We optimize thresholds using grid search on validation data, targeting 2–5 interventions per hour during social periods—a frequency designed to provide meaningful support without becoming intrusive—while minimizing false positives.

The final HASE detection integrates both models through a probabilistic AND operation, where P denotes probability, and θ represents the optimized threshold values:

$$\text{HASE} = (P(S_{\text{arousal}}) > \theta_{\text{arousal}}) \wedge (P(C_{\text{social}}) > \theta_{\text{social}})$$

Intervention Generation and Delivery. Upon HASE detection, the system generates supportive, non-directive prompts delivered through smart glasses. The intervention generation process focuses on universal cognitive and physiological self-regulation techniques such as attention to breathing, momentary pause recommendations, or gentle reminders to consider alternative perspectives.

The intervention delivery mechanism leverages smart glasses to provide discreet support that maintains the natural flow of social interaction. Rather than requiring users to look down at phones or watches, the system delivers subtle visual cues—such as small icons gently fading into the periphery of vision or brief prompts like "Take a deep breath and pause for a moment" or "You might find it helpful to slow down"—that can be easily integrated into ongoing conversations without causing embarrassment or drawing attention from conversation partners.

The system analyzes conversation flow using audio energy patterns to identify natural speaking pauses for appropriate intervention timing. In our initial implementation, interventions are selected from a curated set of static templates to ensure reliability and speed. These templates are pre-generated offline with the assistance of a Large Language Model to ensure they are phrased in a supportive, neutral, and effective manner. Future versions will integrate a lightweight Large Language Model (LLM) to dynamically generate culturally-neutral prompts based on the real-time context vector V_{state} .

2.4 Computational Optimization

Our optimization approach includes post-training quantization (FP32 to INT8) targeting 4× memory reduction. We acknowledge that directly using the complete ImageBind encoder is computationally inefficient for mobile deployment. Therefore, our final solution employs knowledge distillation to transfer the knowledge from the complete ImageBind (as the teacher model) to a lightweight student model. This student model maintains the semantic understanding capabilities while meeting the efficiency requirements for edge deployment. We employ hierarchical processing with lightweight screening followed by full analysis only for potential positive cases.

2.5 Technical Validation Strategy

As the next phase of our work, we will execute the following validation strategy to assess the performance of our framework. The results from this evaluation will form a key component of our final project demonstration. Our strategy is as follows:

- Component-level analysis: C_{social} leverages well-validated sensor classification approaches using the established ExtraSensory dataset
- Cross-modal alignment: The feasibility of our S_{arousal} model relies on ImageBind's demonstrated capabilities in cross-modal representation learning. Therefore, our technical validation strategy includes quantitatively correlating its inference results with ground-truth arousal annotations from

both the WESAD and GLOBEM datasets. This dual-dataset validation provides empirical evidence across both controlled laboratory conditions (WESAD) and real-world longitudinal data (GLOBEM), demonstrating the robustness of our zero-shot approach.

- System integration: The dual-model architecture provides a modular approach that could be validated incrementally
- Computational requirements: We analyze the theoretical computational footprint for mobile deployment through model optimization techniques

3 User Benefits and Applications

Our SocialCompass framework, developed as a proof-of-concept, explores how ubiquitous sensing and foundation models might support users in challenging social situations. For our demonstration, we focus on showcasing the core HASE detection (Phase 3) and smart glasses intervention delivery. The following application scenarios represent the future use cases that this core technology enables and that warrant further investigation:

Supporting Social Communication. For individuals with social anxiety or autism spectrum conditions, our approach explores the feasibility of providing real-time, objective feedback on physiological responses. The proposed smart glasses delivery mechanism aims to offer discreet support that could potentially reduce social avoidance and increase confidence without disrupting the natural flow of conversation.

Enhancing Professional Communication. In communication-intensive professions like healthcare, education, or customer service, managing stress is critical. This work investigates whether a system like SocialCompass could support professionals in maintaining composure during difficult interactions. The unobtrusive nature of the intervention is designed to address the practical need of maintaining professionalism while receiving support.

Fostering Metacognitive Awareness. For general users, the framework could serve as a tool for developing metacognitive awareness. We hypothesize that by receiving gentle cues about their arousal states, users might develop a better understanding of their own emotional and physiological patterns in social contexts, potentially leading to improved communication choices and interpersonal relationships.

4 Ethical Considerations and Responsible Development

Our proof-of-concept framework raises important ethical considerations that must be carefully addressed before any real-world deployment. As an exploration of the intersection between ubiquitous sensing and emotion recognition, we acknowledge the significant responsibility that comes with developing systems capable of monitoring physiological arousal in social contexts. Our preliminary design prioritizes user privacy through an on-device processing architecture, ensuring that raw sensor data remains on the user's personal device. However, we recognize that privacy protection extends beyond technical implementation to include comprehensive user consent protocols, transparent system operation, and robust safeguards against potential misuse.

The risk of behavioral manipulation through real-time arousal detection represents a critical concern that requires careful intervention design. Our exploratory approach emphasizes culturally-neutral, autonomy-preserving interventions that aim to support rather than direct user behavior. We propose maintaining transparent intervention logic and ensuring users retain complete control over system activation and response preferences. For users with clinical conditions or heightened vulnerability, we strongly recommend integration with mental health professionals who can provide appropriate clinical oversight. This exploratory framework allows us to investigate these design principles, but extensive clinical validation and ethical review would be essential before considering any deployment beyond research settings.

Algorithmic bias presents significant concerns, particularly regarding differential responses across demographic groups or cultural backgrounds. We acknowledge this as a fundamental limitation of our current proof-of-concept approach. The ImageBind foundation model and our training datasets may not adequately represent diverse populations, potentially leading to unfair or ineffective responses for certain user groups. We identify the collection of diverse training datasets spanning multiple demographics, cultures, and social contexts as crucial future work to ensure fairness and effectiveness across all user populations. Additionally, our zero-shot inference approach, while technically innovative, remains largely untested across different cultural expressions of stress and arousal, highlighting the need for comprehensive validation studies before any claims of universal applicability can be made.

5 Anticipated Challenges and Limitations

Our SocialCompass framework faces several technical and practical challenges that must be addressed for robust implementation:

Technical Implementation Challenges. Smartphone sensor variability across devices presents a fundamental challenge requiring adaptive calibration protocols. While our preliminary exploration using ImageBind’s sensor encoders provides a starting point, extensive testing across diverse hardware configurations remains necessary. Mobile resource constraints present another limitation—achieving real-time performance while maintaining detection accuracy requires further optimization beyond our proposed knowledge distillation approach. Additionally, distinguishing social stress from other high-arousal states (excitement, physical exercise) remains challenging, though our dual-classification framework represents an initial attempt to improve specificity.

User Acceptance and Deployment Barriers. Intervention timing during social interactions requires careful consideration, as inappropriate prompts could potentially exacerbate rather than alleviate stress. Our conversation flow analysis represents a preliminary approach, but comprehensive user testing across diverse social scenarios would be essential for validation. Despite emphasizing local data processing and user control, individuals may remain skeptical about emotion monitoring systems, necessitating transparent communication about system capabilities and limitations.

Current Approach Limitations. This proof-of-concept investigation acknowledges several constraints on immediate applicability. Our zero-shot arousal inference method requires empirical validation against established psychological measures. Training data for

the social context classifier relies on user-reported labels from ExtraSensory, which may not capture the nuanced nature of social interactions across different cultural contexts. Current intervention strategies represent simplified approaches that would require clinical validation and personalization before real-world deployment.

6 Conclusion

The SocialCompass framework presents a preliminary exploration of how ubiquitous sensors and foundation models might be leveraged for detecting high-arousal social interactions. This proof-of-concept demonstrates the potential of integrating ImageBind’s zero-shot capabilities with ubiquitous sensing to create emotion-aware systems that enhance social interactions while maintaining user privacy and autonomy. The three-phase strategy we propose provides a structured approach for future research in this domain.

While our work establishes an initial technical foundation, we acknowledge that this investigation raises more questions than it answers, particularly regarding the real-world effectiveness of zero-shot arousal inference and its cross-cultural validity. These challenges represent valuable directions for continued research, where more extensive validation and clinical collaboration can further our understanding of how foundation models might support human emotional well-being.

Acknowledgments

We are grateful to Ahmed Farooq for his valuable suggestions and insights.

References

- [1] Rummana Bari, Md Mahbubur Rahman, Nazir Saleheen, Megan Battles Parsons, Eugene H Buder, and Santosh Kumar. 2020. Automated detection of stressful conversations using wearable physiological and inertial sensors. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 4, 4 (2020), 1–23.
- [2] George Boateng. 2020. Towards real-time multimodal emotion recognition among couples. In *Proceedings of the 2020 International Conference on Multimodal Interaction*. 748–753.
- [3] Chulhong Min, Alessandro Montanari, Akhil Mathur, Seungchul Lee, and Fahim Kawsar. 2018. Cross-modal approach for conversational well-being monitoring with multi-sensory earables. In *Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers*. 706–709.
- [4] Leonard C Orji and Umoh A Ita. 2024. AMYGDALA HIJACK: CONTEMPORARY INSIGHTS INTO CAUSES, CORRELATES AND CONSEQUENCES. *Midwifery* 7, 3 (2024), 102–111.
- [5] Philip Schmidt, Attila Reiss, Robert Duerichen, Claus Marberger, and Kristof Van Laerhoven. 2018. Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection. In *Proceedings of the 20th ACM International Conference on Multimodal Interaction* (Boulder, CO, USA) (ICMI '18). Association for Computing Machinery, New York, NY, USA, 400–408. doi:10.1145/3242969.3242985
- [6] Yonatan Vaizman, Katherine Ellis, and Gert Lanckriet. 2017. Recognizing Detailed Human Context in the Wild from Smartphones and Smartwatches. *IEEE Pervasive Computing* 16, 4 (2017), 62–74. doi:10.1109/MPRV.2017.3971131
- [7] Xuhai Xu, Han Zhang, Yasaman Sefidgar, Yiyi Ren, Xin Liu, Woosuk Seo, Jennifer Brown, Kevin Kuehn, Mike Merrill, Paula Nurius, et al. 2022. GLOBEM dataset: multi-year datasets for longitudinal human behavior modeling generalization. *Advances in neural information processing systems* 35 (2022), 24655–24692.